

No. of Functional Features Included in the Solution

Date	07 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview

Smart Lender AI is a Machine Learning-based web application designed to predict loan approval eligibility using applicant information. The solution combines AI-powered prediction, interactive dashboards, EMI calculation, PDF report generation, and a responsive web interface to simplify and accelerate the loan approval process.

Functional Features Table

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Loan Eligibility Prediction	Predicts whether a loan will be approved using a trained Random Forest model.	Machine Learning Engine	Done	High
2	Applicant Information Form	Collects applicant details such as income, education, loan amount, and credit history.	Frontend (HTML/CSS/JS)	Done	High
3	Data Preprocessing & Feature Scaling	Cleans and scales input data before prediction for improved model accuracy.	Backend / ML	Done	High
4	EMI Calculator	Calculates the estimated monthly loan installment based on user inputs.	JavaScript Module	Done	Medium
5	Interactive Dashboard	Displays prediction statistics, model accuracy, and loan analytics using charts.	Dashboard Module	Done	High
6	PDF Report Generation	Generates a downloadable PDF containing applicant details and prediction results.	Report Module	Done	Medium
7	Responsive User Interface	Provides a mobile-friendly, responsive, and user-friendly interface with dark/light mode support.	Frontend UI	Done	Medium

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
8	Cloud Deployment	Deploys the application on Render for online access and scalability.	Deployment	Done	High

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Loan Eligibility Prediction, Data Preprocessing, Applicant Form, Interactive Dashboard, Cloud Deployment

Additional / Nice-to-Have Features

- EMI Calculator
- PDF Report Generation
- Applicant Summary Card
- Dark/Light Theme
- Interactive Charts
- Responsive Design
- AI-based Loan Prediction Dashboard

Features Tested & Verified

- Loan Prediction Accuracy
- Applicant Form Validation
- EMI Calculation
- Dashboard Charts
- PDF Report Download
- Responsive UI
- Render Deployment
- Model Prediction Workflow

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Home Page, Prediction Form, Dashboard, Responsive Design
2	Backend / Logic	4	Flask Routing, Random Forest Prediction, Feature Scaling, EMI Calculation
3	Database / Storage	3	Loan Dataset (CSV), Trained Model (.pkl), Scaler (.pkl)
4	API / Integration	3	Chart.js, PDF Generator, Render Deployment
5	Security / Authentication	2	Input Validation, Secure Session Handling